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October 7, 1968
SUPERSEDING
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December 21, 1956

FEDERAL SPECIFICATION

SIDING (SHINGLES, CLAPBOARDS, AND SHEETS) ASBESTOS-CEMENT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers asbestos-cement siding (shingles, clapboards, and sheets), including accessories.

1.2 Classification.

1.2.1 Types. Asbestos-cement siding shall be of the following types as specified (see 6.2):

Type I - Shingles.
Type II - Clapboards.
Type III - Sheets.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71 - Pallets, Material Handling, Woods, Double Faced, Stringer Construction.
QQ-S-781 - Steel, Strapping, Flat.
QQ-S-790 - Steel Strapping, Round (Bare and Zinc-Coated).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-3448 - Wallboard: Packaging of.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials Standards:

C223 - Standard Specifications for Asbestos-Cement Siding.
C459 - Methods of Sampling and Testing Asbestos-Cement Flat Sheets, Roofing and Siding Shingles, and Clapboards.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P Street N.W., Washingtion, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The asbestos-cement siding, including nails and backer strips, shall be in conformance to ASTM C223, and as specified herein.

3.2 Size. Sizes shall be as specified (see 6.2), and shall conform to ASTM C223.

3.3 Color. Unless otherwise specified (see 6.2), the color of the exposed surface of the siding shall be natural cement gray.

3.4 Finish. The exposed surface of the siding shall be smooth, grained, coated, or textured as specified (see 6.2).

3.5 Holes for nails and fasteners. Unless otherwise specified (see 6.2), holes for nails and fasteners on type I shingles and type II clapboards shall be provided to satisfy the top lap requirements. When specified (see 6.2), holes for nails and fasteners shall be provided on type III sheets. Spacing of holes shall be as specified (see 6.2).

3.6 Special fasteners. When specified (see 6.2), special fasteners for attaching siding units to other than a wood lumber nailing base shall be furnished. Special fasteners shall be of nonstaining, corrosion-resistant type.

3.7 Workmanship. The surface of the siding to be exposed shall be free of cracks, breaks, or other defects.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be by individual shingles, clapboards, or sheets, as applicable, and shall be in accordance with MIL-STD-105.

4.3.1.2 For tests. Samples for tests shall be in accordance with ASTM C459.

4.3.2 Examination. Samples selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1.2 shall be tested in accordance with ASTM C459. Failure of any test shall be cause for rejection of the lot.

4.4 Inspection procedure.

4.4.1 Examination. The siding shall be examined for the following defects:

101. Type not as specified.
102. Dimensions not as specified.
103. Finish not as specified.
104. Color not as specified.
105. Holes for nails and fasteners not as specified.
106. Nails and backing strips not provided as required.
107. Workmanship not as specified.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108. Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall constitute one defect.
109. Bundling not as specified.
110. Wire strapping not as specified.
111. Packing not as specified.
112. Strapping not zinc-coated for level A.
113. Marking illegible, incorrect, incomplete, or missing.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Bundling. Type I shingles and type II clapboards shall be bundled as specified in 5.1.1.1.1 and 5.1.1.1.2. Type III sheets shall not require bundling.

5.1.1.1.1 Type I, shingles. Shingles of like size and description shall be bundled with the number of shingles per bundle equally divided so that multiple bundles will equal one square. The bundles shall be secured with two, round wire, 18-gauge steel straps conforming to QQ-S-790, finish 2.

5.1.1.1.2 Type II, clapboards. Clapboards of like size and description shall be in bundles up to 100 pounds maximum weight (plus or minus 10 percent). The bundles shall be secured with two or more round wire, 18-gauge steel straps conforming to QQ-S-790, finish 2, spaced not more than 24 inches apart and located no closer than 3 inches nor farther than 6 inches from each end.

5.1.2 Level C. The siding shall be packaged in accordance with the suppliers's standard practice.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I, shingles. Singles of like size and description, bundles as specified in 5.1.1.1.1, shall be packed in wood, cleated-plywood boxes conforming to PPP-B-601, overseas type, style A or B, type 1 load. The boxed shingles shall be in multiples of squares, up to the weight limitation of the container. Strapping shall be zinc-coated and shall conform to QQ-S-781.

5.2.1.2 Type II, clapboards. Clapboards of like size and description, bundled as specified in 5.1.1.1.2, shall be palletized or crated as follows: Clapboards up to and including 84 inches shall be palletized. Pallets shall conform to NN-P-71, type II, except that pallet dimensions are to accommodate the length of clapboards (pallet width up to 84 inches). Pallet length for all 12-inch clapboards shall be 48 inches. For clapboards less than 12-inch width, pallet length shall be reduced, as required, to equal the combined widths of the bundled clapboards per layer. For pallet widths over 60 inches, five stringers shall be required. Clapboards longer than 84 inches shall be packed in a crate conforming to MIL-C-3774, nailed assembly, skid-type base. Blocking, bracing and anchoring of the bundled clapboards within the crate shall be in accordance with MIL-STD-1186. Strapping shall conform to QQ-S-781, zinc-coated.

5.2.1.3 Type III, sheets. The asbestos-cement sheets shall be packed in accordance with MIL-W-3448.

5.2.2 Level C. Shingles, clapboards, and sheets, with the shingles and clapboards bundled as specified in 5.1.1.1, shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rates in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with FED. STD. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The asbestos-cement siding is intended to be used for the weather-exposed sidewall surfaces of buildings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2.1).
(c) Size required (see 3.2).
(d) Color required when other than natural cement gray (see 3.3).

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(e) Finish required (see 3.4).
(f) When holes for nails of special fasteners on type I and type II siding are not required (see 3.5).
(g) When holes for nails or special fasteners on type III sheets are required, and the number and spacing of the holes required (see 3.5).
(h) When special fasteners are required (see 3.6).
(i) Level of packaging and level of packing required (see 5.1 and 5.2).

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Air Force - 85

User activity:

Army - CE

Preparing activity:

Army - ME

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